

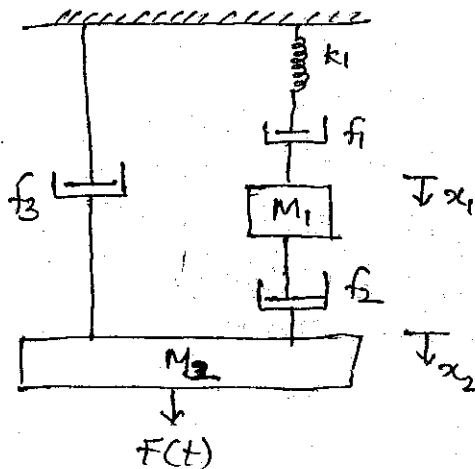
B. Tech Degree VI Semester Examination, April 2009

ME 603 INSTRUMENTATION THEORY AND CONTROL ENGINEERING (2002 Scheme)

Time : 3 Hours

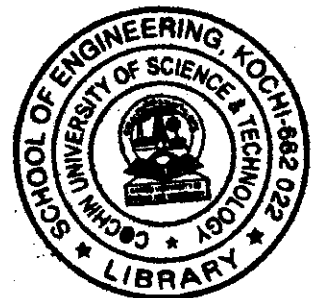
Maximum Marks : 100

- I. (a) Explain *any three* important characteristics to be considered in selecting a measuring instrument. (12)
 (b) Define Linearity and Repeatability with reference to a measurement system. (8)
- OR**
- II. (a) With the help of a neat block diagram, explain the functional elements of a measurement system with pressure thermometer as an example. (12)
 (b) What is the relationship between sensitivity and range? What is the disadvantage of very sensitive instrument? (8)
- III. (a) Explain any one type of force measuring technique. (8)
 (b) Explain the working principle and application of a Geiger Muller Counter. (12)
- OR**
- IV. Write short notes on the following :
 (i) Infrared pyrometry (7)
 (ii) ORSAT's apparatus (7)
 (iii) Pneumatic load cell (6)
- V. Obtain the mathematical model for the mechanical system given below. Also obtain the mechanical circuit diagram using force-current and force-voltage analogy. (20)



OR

(Turn Over)



- VI. (a) Evaluate the dynamic error coefficients for the unity feedback system with forward path transfer function $G(s) = \frac{12}{s^2(s+3)}$. (12)
- (b) Obtain the state space representation of a series RLC circuit. (8)
- VII. (a) Using Hurwitz approach, verify the stability of $s^5 + 4s^4 + 9s^2 + 8s + 2 = 0$ (10)
- (b) What is Liapunov function? How it is being developed? Explain how it can be used in stability analysis of the system. (10)
- OR**
- VIII. Sketch the Bode plot for the open loop transfer function for the unity feedback system given below and assess stability $G(s) = \frac{15}{s(s+3)}$. Determine its gain margin and phase margin. (20)
- IX. (a) Write down the transfer function of an AC servomotor. (10)
- (b) Explain the constructional details and characteristics of tachogenerator. (10)
- OR**
- X. Write short notes on the following :
- (i) Pneumatic controllers (6)
 - (ii) PID Control (8)
 - (iii) DC Servomotor (6)
